

A10-RPT: Mission-Variable-Preserving Structured Priors for Rocket-Propulsion Surrogates under Thermal, Supply, and Combined Stress

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Abstract

This paper introduces A10-RPT, a mission-variable-preserving structured-prior control framework for reduced rocket-propulsion surrogates under thermal, supply-margin, growth-rate, noise/disturbance, and combined stress. The objective is not to propose a new propulsion mechanism, engine geometry, propellant system, or hardware-ready controller. Instead, a closed nondimensional surrogate is constructed with chamber-pressure, pressure-oscillation, thermal-load, combustion-phase, supply-margin, and cumulative-impulse states. Feasibility-first controllers are evaluated by failure rate, unsafe duration, and CVaR_{0.9} of safe-set violation. An initial A10 controller reduced unsafe excursions but depleted supply margin. A monolithic supply-aware gate failed to resolve this bottleneck, whereas a separated supply barrier eliminated supply-margin failures in the nominal ensemble. A subsequent thermal barrier achieved complete in-distribution mission feasibility in the tested ensemble. Stress validation then revealed that combined regimes are not purely controller-limited. Resource-frontier sweeps showed that moderate combined stress becomes feasible at a supply-resource scale of approximately 1.50–1.55, high thermal stress becomes near-feasible around cooling scale 2.50 with rare residual thermal failures, and harsh combined stress remains a severe thermal-limited near-frontier case under the tested resource family. These results support A10-RPT as a feasibility-diagnostic control theory: it separates controller-feasible, resource-frontier-feasible, near-frontier, and architecture-limited regimes in a reduced propulsion surrogate.

Scope and safety note. This work is a nondimensional surrogate-control study. It is not an engine-design tool, not a propulsion-performance predictor, and not a prescription for flight hardware, hazardous experiments, propellant handling, or real engine operation. All variables, resource scales, and controllers in this manuscript are abstract quantities for theoretical validation only.

1 Introduction

Rocket propulsion systems are governed by hard constraints: chamber pressure, combustion instability, wall heat load, supply-system margin, actuator-rate limits, and required impulse must be satisfied simultaneously [1, 2]. In such systems, maximizing nominal performance is not equivalent to mission feasibility. A controller that produces high impulse can still fail if it depletes supply margin, excites an unstable mode, or accumulates thermal load beyond a hard boundary.

This paper studies a reduced feasibility-first control problem: can a structured prior preserve mission variables while reducing thermal, supply, and combined-stress failures in a closed rocket-propulsion surrogate? The proposed framework, A10-RPT, is not a new propulsion principle. It is a structured-prior control architecture that gates corrective actions near safety boundaries, separates newly exposed bottlenecks into explicit barriers, and diagnoses whether a stress regime is controller-feasible or resource-limited.

The main contributions are as follows:

1. a nondimensional rocket-propulsion surrogate with pressure, oscillation, thermal, phase, supply-margin, and impulse states;
2. a hierarchy of A10-style structured-prior controllers and barrier separations;
3. stress validation under thermal, growth-rate, supply, noise/disturbance, and combined regimes;
4. resource-frontier maps identifying controller-feasible, resource-frontier-feasible, near-frontier, and unresolved architecture-limited regimes.

The approach is related in spirit to safe-set and viability-based thinking, set invariance, control-barrier methods, and risk-sensitive metrics such as CVaR [4, 3, 5, 6], but the present manuscript does not claim formal barrier-certificate guarantees. The objective is more modest: to build an interpretable, closed, intermediate surrogate in which feasibility-oriented controller structure and resource frontiers can be tested without pretending to model a flight engine.

2 Reduced Rocket-Propulsion Surrogate

2.1 State and control variables

Let nondimensional burn time be $\tau \in [0, 1]$. The state is

$$x(\tau) = \begin{bmatrix} p & a & h & \varphi & m & I \end{bmatrix}^\top, \quad (1)$$

where p is a chamber-pressure proxy, a is pressure-oscillation amplitude, h is a thermal-load proxy, φ is a combustion phase/composition response variable, m is supply-system margin, and I is cumulative impulse attainment. The control vector is

$$u(\tau) = \begin{bmatrix} u_P & u_\varphi & u_C & u_D \end{bmatrix}^\top, \quad (2)$$

representing power/throttle, phase/composition, cooling, and damping channels.

Table 1: State variables and primary safety interpretation. The variables are nondimensional surrogate states, not direct measurements from a particular propulsion system.

Variable	Meaning	Safety interpretation
p	Chamber-pressure proxy	$p \leq 1$
a	Pressure-oscillation amplitude	$a \leq 1$
h	Thermal-load proxy	$h \leq 1$
φ	Combustion phase/composition response state	$ \varphi \leq 1$
m	Supply-system margin	$m \geq m_{\min}$
I	Cumulative impulse attainment	$I(1) \geq 1$

Table 2: Control channels used by the reduced surrogate. Each channel represents an abstract authority class rather than a hardware prescription.

Channel	Abstract role	Interpretation in the surrogate
u_P	Power/throttle-like authority	Raises or lowers chamber-pressure and impulse generation.
u_φ	Phase/composition authority	Changes the combustion response variable and can reduce or increase instability depending on state.
u_C	Cooling authority	Reduces accumulated thermal load and indirectly damps instability.
u_D	Damping authority	Acts directly on the pressure-oscillation mode.

2.2 Dynamics

The surrogate dynamics are

$$\dot{p} = \frac{1}{\tau_p} [p_{\text{cmd}}(u_P, \varphi) - p] + \eta_a a^2 + \xi_p, \quad (3)$$

$$p_{\text{cmd}} = p_0 + k_P u_P + k_\varphi \varphi, \quad (4)$$

$$\dot{a} = [\sigma_0 + \sigma_p(p - p_s) + \sigma_\varphi \varphi - \sigma_D u_D - \sigma_C u_C] a - \kappa_a a^3 + d_a(\tau), \quad (5)$$

$$\dot{h} = \frac{1}{\tau_h} [q(p, a, \varphi) - h] - \gamma_C u_C + \xi_h, \quad (6)$$

$$q(p, a, \varphi) = q_0 + q_p p^2 + q_a a^2 + q_\varphi |\varphi|, \quad (7)$$

$$\dot{\varphi} = -\frac{1}{\tau_\varphi} \varphi + b_\varphi u_\varphi + c_p(p - p_s) + \xi_\varphi, \quad (8)$$

$$\dot{m} = -\lambda_P u_P^2 - \lambda_{\dot{P}} \dot{u}_P^2 - \lambda_\varphi u_\varphi^2 + \rho_m(1 - m), \quad (9)$$

$$\dot{I} = \chi_I [p - c_\varphi \varphi^2 - c_a a^2]_+. \quad (10)$$

This form intentionally captures only closed intermediate constraints rather than high-fidelity combustion physics, following the spirit of reduced instability and propulsion-system abstractions rather than detailed engine modeling [1, 2]. The pressure-oscillation state a is driven by an effective growth-rate term and nonlinear saturation; the thermal state h accumulates pressure, oscillation, and phase-response contributions; and the supply-margin state m penalizes both sustained high power and rapid power changes. Control-input saturation and slew-rate limits are applied in all simulations.

2.3 Safe set and mission condition

Consistent with viability and set-invariance terminology [4, 3], the safe set is

$$\mathcal{S}_{\text{RPT}} = \{x : p \leq 1, a \leq 1, h \leq 1, |\varphi| \leq 1, m \geq m_{\min}\}. \quad (11)$$

A trajectory succeeds if

$$x(\tau) \in \mathcal{S}_{\text{RPT}} \quad \forall \tau \in [0, 1], \quad I(1) \geq 1. \quad (12)$$

The experiments in this manuscript use $m_{\min} = 0.15$ unless otherwise noted.

3 Metrics and Stress Protocol

The primary metrics are failure rate, mean unsafe duration, CVaR_{0.9} of safe-set violation V , final impulse, minimum final impulse, and failure-reason decomposition into pressure, oscillation, thermal, supply-margin, and impulse components. The safe-set metrics follow the viability and invariance viewpoint, while CVaR is used as a tail-risk summary rather than as a proof of stochastic optimality [4, 3, 6]. The violation integral is

$$V = \int_0^1 \left[(p-1)_+^2 + (a-1)_+^2 + (h-1)_+^2 + (|\varphi|-1)_+^2 + (m_{\min} - m)_+^2 \right] d\tau. \quad (13)$$

Failure rate is defined as the fraction of ensemble trajectories with either a safe-set violation or $I(1) < 1$. Unsafe duration is the fraction of nondimensional burn time spent outside $\mathcal{S}_{\text{RPT}}$.

Stress profiles include nominal retest, growth-rate stress, heat stress, supply stress, noise/disturbance stress, combined-moderate stress, and combined-harsh stress. The stress profiles do not represent a specific engine environment. They are nondimensional stress tests used to determine whether a controller is operating inside a feasible authority envelope or merely exploiting nominal conditions.

4 A10-RPT Barrier Architecture

The A10 controller begins with a danger-gated structured prior,

$$u(\tau) = u_{\text{nom}}(\tau) + g_A(D)\Delta u_A(x, \dot{x}, \tau), \quad (14)$$

where $g_A(D)$ activates smoothly near safety boundaries. In the early versions, the main danger gate was asked to handle all constraints simultaneously. The later versions separated dominant bottlenecks into explicit barriers.

Table 3: Ablation history of the A10-RPT controller family. The progression is intentionally treated as a diagnostic sequence rather than as a claim that one controller is universally optimal.

Version	Main change	Diagnostic outcome
v0.1	Initial A10 danger-gated prior	Reduced unsafe excursions but depleted supply margin.
v0.2	Monolithic inclusion of supply margin in the main danger gate	Degraded feasibility; the supply bottleneck could not simply be mixed into the main gate.
v0.3	Separated supply barrier	Eliminated supply-margin failures in the nominal ensemble but left thermal failures.
v0.4	Thermal-polished barrier	Achieved complete in-distribution feasibility in the tested ensemble.
v0.6	Adaptive and supply-priority variants	Improved supply and noise stress relative to v0.4.
v0.7	Minimax thermal-stress specialist	Improved heat-stress and combined-regime tail risk.
v0.8–v0.10	Architecture/resource-frontier sweeps	Identified resource scales required to open or approach feasible regions.

The central architectural lesson is that A10-RPT is not a single universal gate. It is a feasibility-first structured prior whose barriers are separated and reweighted as the dominant bottleneck migrates.

5 Results I: In-Distribution Feasibility

The v0.4 thermal-polished controller achieved complete feasibility in the tested in-distribution ensemble: failure rate 0, unsafe duration 0, and $\text{CVaR}_{0.9}(V) = 0$. It retained positive impulse margin and kept the worst thermal and supply states inside the safe set.

Table 4: Representative in-distribution result for the adopted v0.4 controller. Values are from the tested 120-case ensemble.

Controller	Failure	Unsafe duration	$\text{CVaR}_{0.9}(V)$	Mean $I(1)$	Worst h	Worst m
A10 thermal v0.4	0.000	0.000	0.000	1.174129	0.966376	0.260027

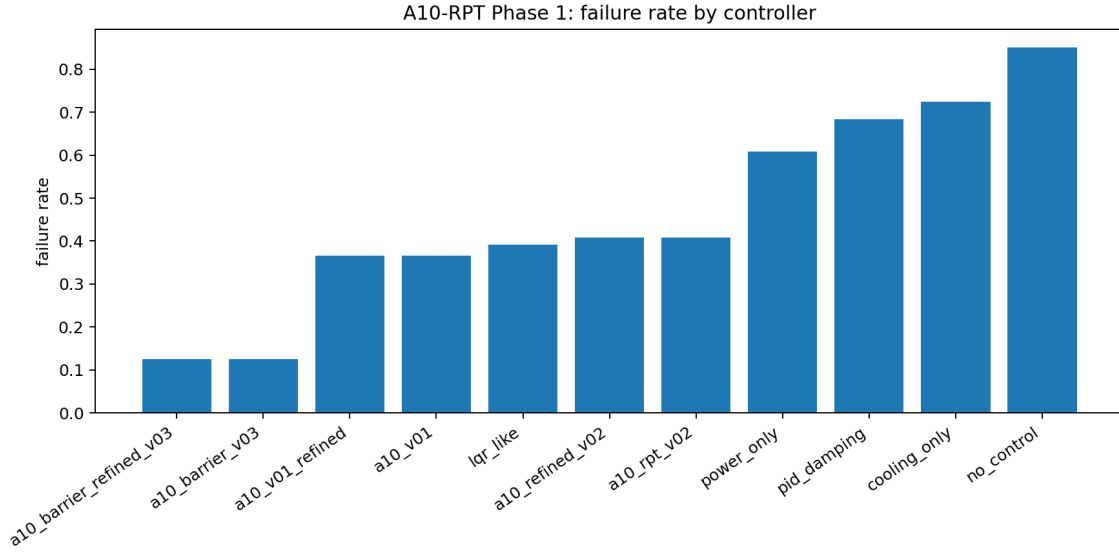


Figure 1: In-distribution failure-rate comparison across controller versions and baselines. The v0.4 thermal-polished controller and its refined variant achieved zero failure rate in the tested nominal ensemble.

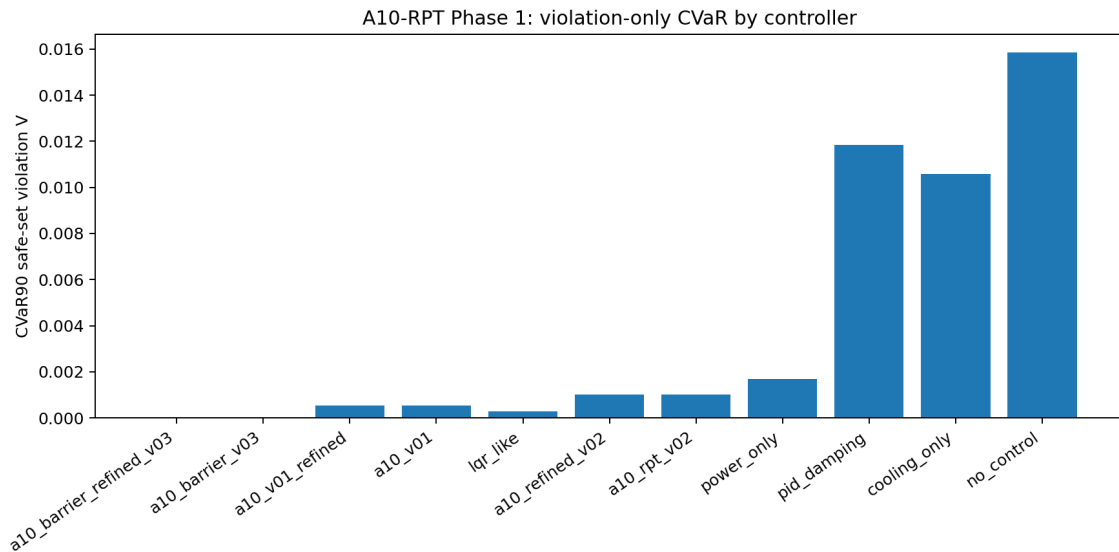


Figure 2: In-distribution $\text{CVaR}_{0.9}$ of safe-set violation. The v0.4 thermal-polished controller eliminated violation tail risk in the tested nominal ensemble.

6 Results II: Stress Validation and Bottleneck Migration

Stress validation showed that controller performance depends on the active bottleneck. The v0.4 controller was strong in nominal and growth-stress regimes, but under supply and noise/disturbance stress it could again consume excessive supply margin. Adaptive and supply-priority v0.6 variants improved those regimes, while the v0.7 minimax variant improved heat-stress and combined-regime tail behavior.

The practical interpretation is that controller evolution exposes a moving bottleneck hierarchy:

$$\begin{aligned} &\text{oscillation/thermal risk} \rightarrow \text{supply-margin depletion} \\ &\rightarrow \text{thermal accumulation} \rightarrow \text{severe combined thermal limitation.} \end{aligned} \tag{15}$$

This hierarchy is a central result of the surrogate study. A10-RPT is most useful when it does not merely reduce a fixed scalar loss, but instead identifies which constraint is dominant and whether that constraint can be addressed by controller structure or requires resource expansion.

7 Results III: Resource-Frontier Sweeps

The resource-frontier sweep varies abstract cooling, supply, and impulse-resource scales. These scales are nondimensional authority multipliers. They must not be interpreted as direct hardware mass, turbopump size, cooling-channel area, or propellant-flow multipliers. The purpose is to identify which stress regimes are controller-feasible, resource-frontier-feasible, near-feasible, or outside the tested resource family.

7.1 Combined-moderate stress

High-N confirmation showed that combined-moderate stress becomes feasible under supply-resource expansion. The best confirmed result used $C = 1.00$, $S = 1.55$, and $I_s = 1.00$ with the A10 minimax v0.7 controller, reaching zero failure rate and zero unsafe duration. The minimal boundary is near $S \approx 1.50$, but that setting left rare failures and a supply margin close to the safety boundary. Thus $S \approx 1.55$ is the recommended surrogate frontier.

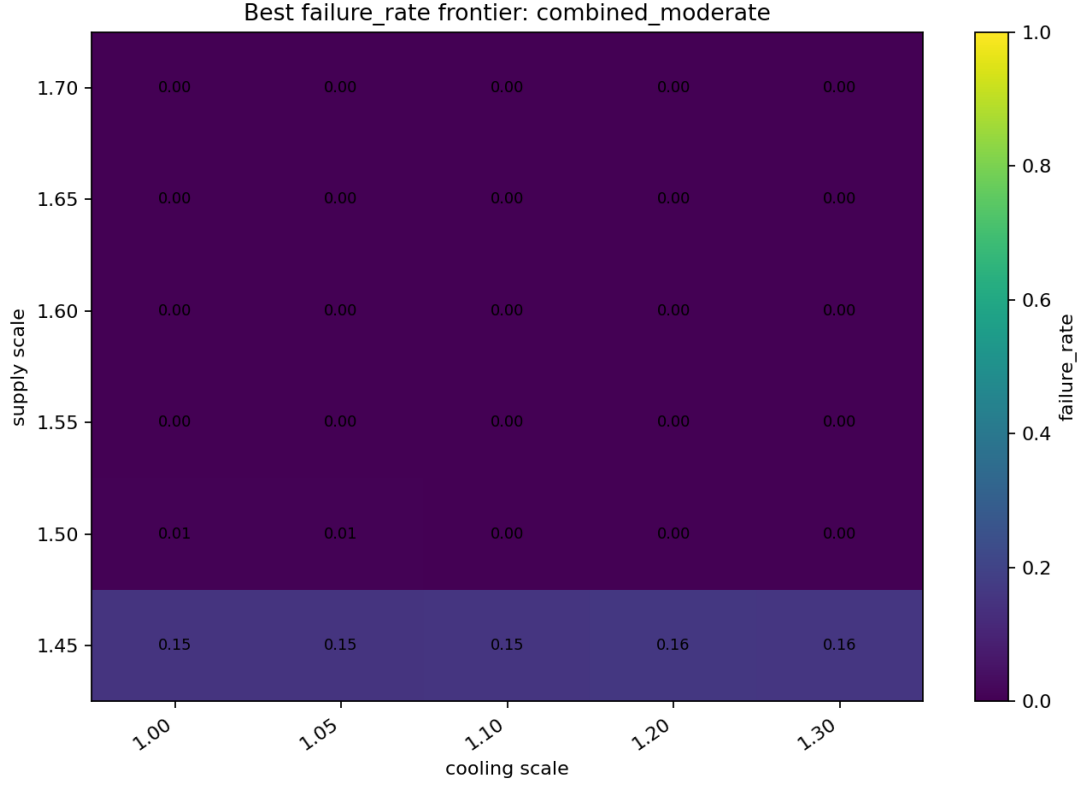


Figure 3: High-N resource-frontier confirmation for the combined-moderate profile. The vertical transition around supply scale 1.50–1.55 indicates a supply-dominated frontier; cooling scale is comparatively secondary in this regime.

7.2 High thermal stress

The heat_150 profile is a high-cooling near-frontier case. At $C = 2.50$, $S = 1.20$, and $I_s = 1.00$, the failure rate was reduced to 0.006667, with remaining failures concentrated in rare thermal exceedances. This is not strict complete feasibility, but it is close enough to classify the regime as high-cooling near-feasible rather than controller-impossible.

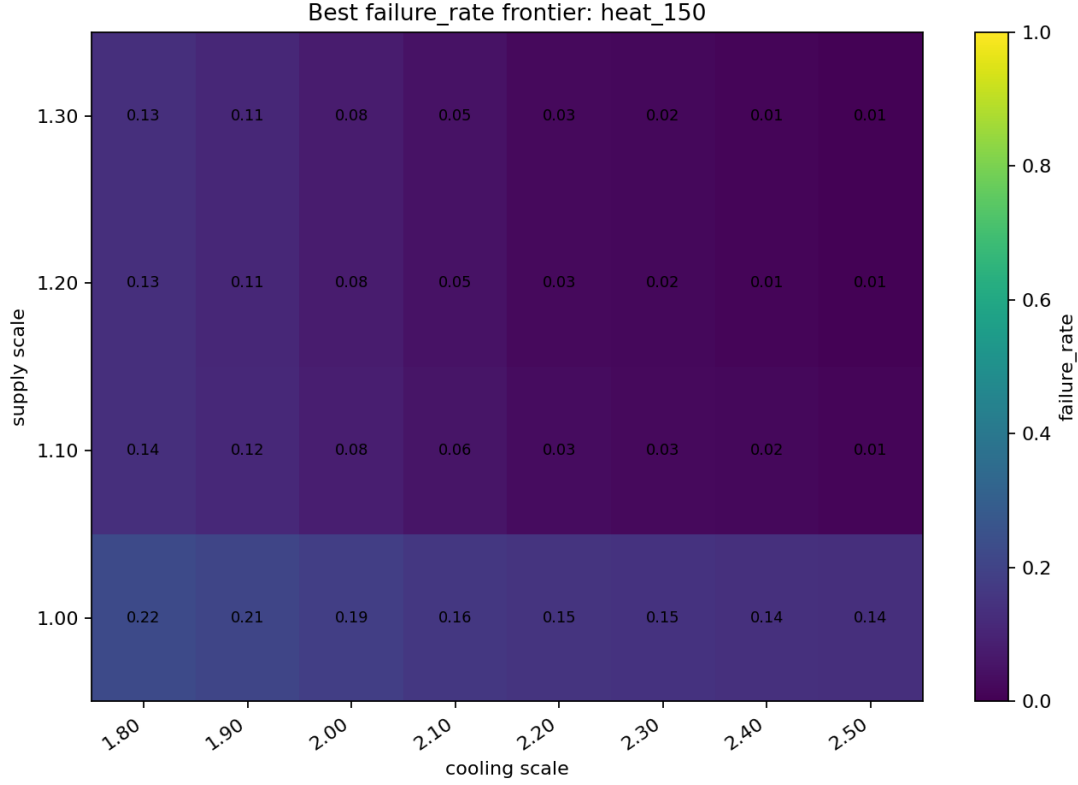


Figure 4: High-N resource-frontier confirmation for heat_150. Failure rate decreases primarily with cooling scale. The best tested region approaches feasibility at cooling scale around 2.50, with small residual thermal exceedance.

7.3 Harsh combined stress

The combined-harsh regime remains unresolved under the tested family, but it is no longer simply infeasible. At $C = 2.50$, $S = 1.75$, and $I_s = 1.10$, the best observed failure rate was 0.075, with residual failures dominated by thermal exceedance. Supply-margin and impulse failures were eliminated at the best point, indicating that the remaining bottleneck is thermal rather than supply or impulse.

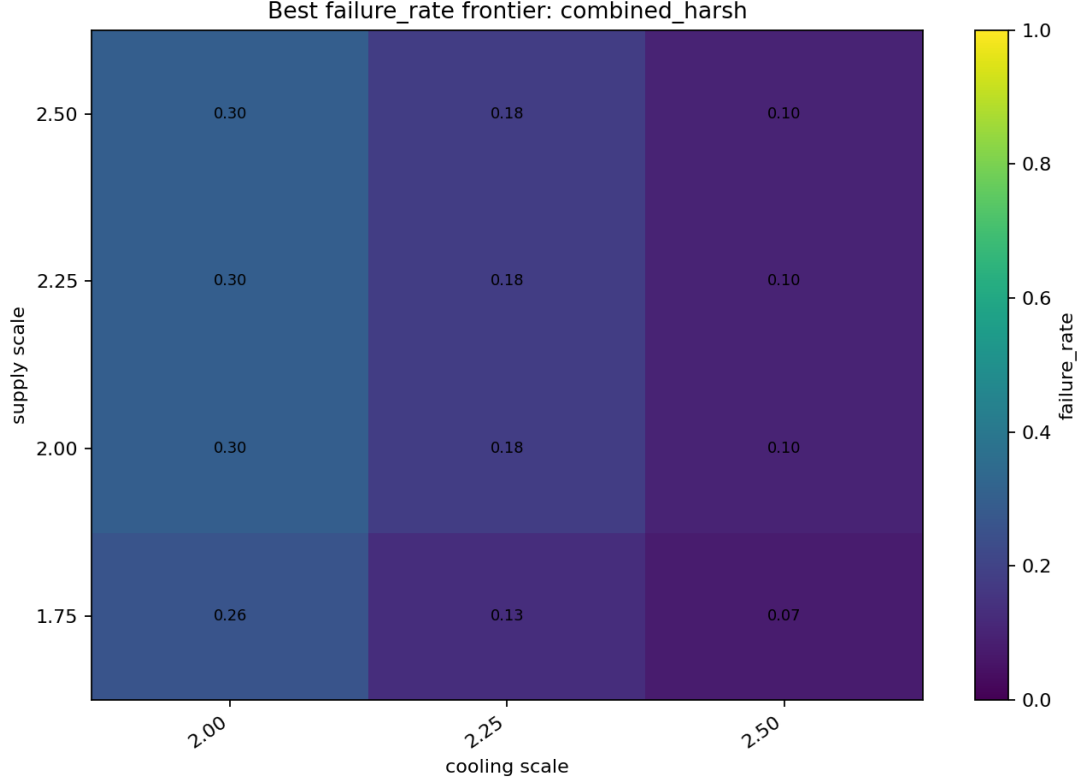


Figure 5: High-N resource-frontier confirmation for the combined-harsh profile. Even after expanding the resource family, the best observed failure rate remains above the target threshold. Residual failures are thermal-dominated, placing this regime in a severe thermal-limited near-frontier class.

Table 5: High-N frontier confirmation summary. C , S , and I_s denote cooling, supply, and impulse-resource scales in the nondimensional surrogate.

Profile	Controller	C	S	I_s	Failure	Unsafe	Worst h	Worst m
combined_moderate	A10 minimax v0.7	1.00	1.55	1.00	0.0000	0.0000	0.975344	0.200770
heat_150	A10 minimax v0.7	2.50	1.20	1.00	0.0067	0.0015	1.008569	0.195385
combined_harsh	A10 minimax v0.7	2.50	1.75	1.10	0.0750	0.0154	1.043346	0.223890

Table 6: Regime classification inferred from the Phase 1 surrogate experiments.

Class	Regime	Interpretation
I	nominal, growth, supply, noise	Controller-feasible under suitable A10 barrier variants.
II	combined_moderate	Supply-resource-frontier feasible; recommended surrogate condition $S \approx 1.55$.
III	heat_150	High-cooling near-feasible thermal frontier; rare residual thermal exceedance remains near $C \approx 2.50$.
IV	combined_harsh	Severe thermal-limited near-frontier under the tested family; unresolved at target failure rate 0.05.

8 Discussion

The results support the interpretation of A10-RPT as a feasibility-diagnostic theory rather than a performance-maximizing controller. Three points are central.

First, the dominant bottleneck migrates. The initial A10 prior reduced unsafe excursions but exposed supply-margin depletion. Supply-barrier separation then exposed thermal accumulation. Resource-frontier expansion finally showed that the harsh combined regime is dominated by residual thermal failure rather than by supply or impulse failure.

Second, a monolithic danger gate is insufficient. Simply mixing every constraint into one scalar danger index degraded feasibility in v0.2. Better results appeared only when the supply and thermal bottlenecks were separated as barriers with their own priority logic.

Third, resource-frontier analysis gives the theory practical diagnostic value. Combined-moderate stress is not solved by controller tuning alone at baseline resource scale, but it opens when supply resource is expanded to approximately 1.50–1.55. Heat_150 approaches feasibility as cooling scale rises to approximately 2.50. Combined-harsh remains a severe thermal-limited near-frontier case in the tested resource family.

9 Limitations

The surrogate is low-dimensional and nondimensional. It is not calibrated against a specific engine, propellant, turbopump, injector, combustion chamber, material system, cooling channel, structural design, sensor suite, or flight controller. Resource scales are abstract authority multipliers and must not be interpreted as direct hardware mass, size, flow-rate, or cooling-channel multipliers. Higher-fidelity thermal, acoustic, fluid, structural, and actuator models are required before any engineering interpretation.

The results also depend on the chosen stress profiles, uncertainty ranges, controller parameterization, and evaluation horizon. Therefore, the present evidence supports a structural feasibility argument within the surrogate, not a universal statement about rocket engines.

10 Conclusion

A10-RPT provides a structured-prior control framework for mission feasibility in a reduced rocket-propulsion surrogate. The experiments show that nominal and single-stress regimes can be controller-feasible, moderate combined stress opens under supply-resource expansion, high thermal stress approaches feasibility under high cooling-resource expansion, and harsh combined stress remains limited by residual thermal failures. The contribution is not a hardware controller but a feasibility-diagnostic theory that identifies where control suffices, where resource expansion opens feasibility, and where deeper architectural change is required.

Reproducibility note

The numerical results reported here were generated with the A10-RPT Phase 1 scripts v0.4–v0.10. The accompanying source CSV files and figures should be preserved with the manuscript package. Because the surrogate is nondimensional, reproducibility means matching the specified scripts, random seeds, stress profiles, and resource multipliers, not matching any physical engine dataset.

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